www.tntextbooks.in

derived from the Gk. *Xylos* – wood. The xylem which is derived from Procambium is called **primary xylem** and the xylem which is derived from vascular cambium is called **secondary xylem**. Early formed primary xylem elements are called protoxylem, whereas the later formed primary xylem elements are called metaxylem.

Protoxylem lies towards the periphery and metaxylem that lies towards the centre is called **Exarch.** It is common in *roots.*

Protoxylem lies towards the centre and meta xylem towards the periphery this condition is called **Endarch**. It is seen in *stems.*

Protoxylem is located in the centre surrounded by the metaxylem is called **Centrarch**. In this type only one vascular strand is developed. Example: *Selaginella sp.*

Protoxylem is located in the centre surrounded by the metaxylem is called **Mesarch**.In this type several vascular strands are developed. Example: *Ophioglossum sp.*

Student Activity

Cell lab: students prepare the slide and identify the different types tissues.

Xylem Consists of Four Types of Cells

1. Tracheids
2. Vessels or Trachea
3. Xylem Parenchyma
4. Xylem Fibres

Xylem is called **hadrome phloem** is called **leptome.** These terms are coined by **haberlandt (1914)**

Tracheids

Tracheids are dead, lignified and elongated cells with tapering ends. Its lumen is broader than that of fibres. In cross section, the tracheids are polygonal.

There are different types of cell wall thickenings due to the deposition of secondary wall substances. They are annular (ring like), spiral (spring like), scalariform (ladder like) reticulate (net like) and pitted (uniformly thick except at pits). Tracheids are imperforated cells with bordered pits on their side walls. Only through this conduction takes place in Gymnosperms. They are arranged one above the other. Tracheids are chief water conducting elements in Gymnosperms and Pteridophytes. They also offer mechanical support to the plants.

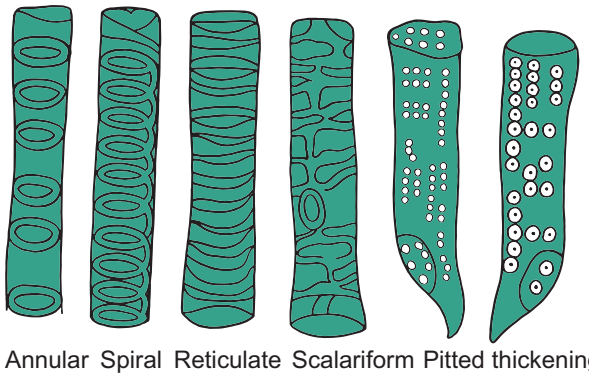


Figure 9.10: Types of secondary wall thickenings in tracheids and vessels

Vessels or Trachea

Vessels are elongated tube like structure. They are dead cells formed from a row of vessel elements placed end to end. They are perforated at the end walls. Their lumen is wider than Tracheids. Due to the dissolution of entire cell wall, a single pore is formed at the perforation plate. It is called **simple perforation plate**, Example: *Mangifera*. If the perforation